



REAL – TIME OCCUPANCY DETECTION SYSTEM FOR ENERGY EFFICIENCY

Using Arduino Uno and Dual IR Sensors

submitted as part of Internship Course (L & T EduTech)

on

Microcontrollers & Applications

by

P SUCHARITA SAI

24BES7051

B. Tech (ECE)

VIT-AP University

Abstract

The Real-Time Occupancy Detection System is an Arduino Uno-based embedded system that monitors the number of people entering and exiting a room using two Infrared (IR) sensors. The Arduino continuously updates the occupancy count and controls an 8-channel relay module based on the number of occupants. The relay module's onboard LEDs provide a visual indication of the occupancy count, while the Serial Monitor displays real-time occupancy information. The prototype demonstrates a simple, reliable, and cost-effective occupancy monitoring solution that can be extended for smart homes, offices, classrooms, and other building automation applications.

1. Introduction

Occupancy detection is an important application of embedded systems that enables intelligent monitoring and automation of indoor environments. Conventional systems rely on manual operation, which often results in electrical appliances remaining switched on even when rooms are unoccupied, leading to unnecessary energy consumption.

This project presents a Real-Time Occupancy Detection System using an Arduino Uno, two Infrared (IR) sensors, and an 8-channel relay module. The system detects entry and exit events, maintains a real-time occupancy count, and demonstrates occupancy-based automation by controlling the relay module. The occupancy count is also displayed on the Arduino Serial Monitor, providing continuous monitoring of room occupancy.

2. Problem Statement

In homes, offices, classrooms, and other indoor environments, electrical appliances are often left switched on even when no occupants are present. This results in unnecessary energy consumption and increased electricity costs. An automated occupancy detection system can overcome this problem by monitoring the number of people entering and leaving a room and enabling occupancy-based automation.

3. Objectives

- Detect people entering and exiting a room using two IR sensors.
- Maintain a real-time occupancy count.
- Control an 8-channel relay module based on the occupancy count.
- Display occupancy information on the Arduino Serial Monitor.
- Develop a simple, reliable, and low-cost occupancy detection system.

4. Hardware Requirements

S. No.	Component	Quantity
1.	Arduino Uno	1
2.	IR Sensors	2
3.	8-Channel Relay Module	1
4.	Breadboard	1
5.	Jumper Wires	As required
6.	USB Cable	1

5. Software Components

S.No.	Software	Purpose
1.	Arduino IDE	Writing and Compiling
2.	Arduino C++	Programming Language

6. Hardware Description

Arduino Uno

The Arduino Uno serves as the main controller of the system. It receives signals from the IR sensors, updates the occupancy count, controls the relay module, and displays the occupancy information on the Serial Monitor.

IR Sensors

Two IR sensors are used to detect entry and exit events. One sensor is placed at the entrance and the other at the exit to update the occupancy count accurately.

8-Channel Relay Module

The relay module demonstrates occupancy-based automation. As the occupancy count increases, additional relay channels are activated, and their onboard LEDs indicate the active channels.

Breadboard

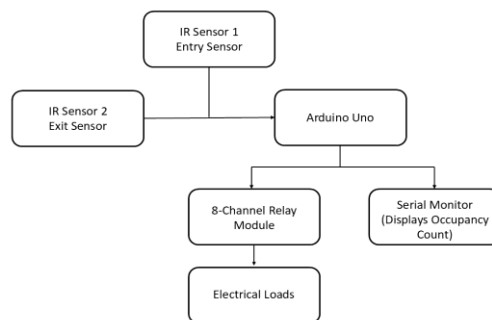
The breadboard provides power distribution and electrical connections between the Arduino Uno, IR sensors, and relay module.

Jumper Wires

Jumper wires are used to establish connections between all hardware components.

7. Block Diagram

The below diagram illustrates the overall architecture of the proposed Real-Time Occupancy Detection System.



The block diagram represents the overall operation of the proposed occupancy detection system. Two IR sensors are placed at the entry and exit points of the room. The Entry IR Sensor detects people entering the room, while the Exit IR Sensor detects people leaving.

Both sensors send digital signals to the Arduino Uno, which acts as the central controller. The Arduino processes these signals, updates the occupancy count, and continuously monitors the current room occupancy.

Based on the occupancy count, the Arduino controls the 8-channel relay module. The onboard LEDs of the relay module provide a visual indication of the occupancy level. Simultaneously, the occupancy count is displayed on the Serial Monitor, allowing real-time monitoring of the system.

8. Working Principle

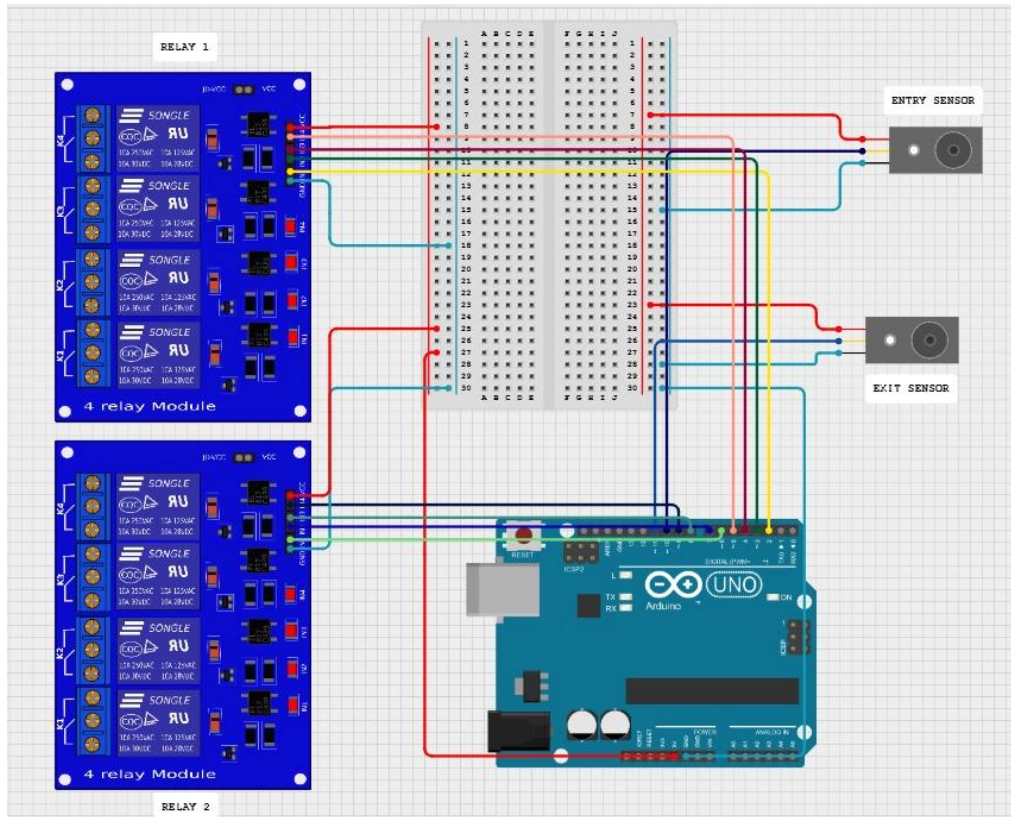
The system uses two Infrared (IR) sensors positioned at the entry and exit points of a room. The Entry IR Sensor detects people entering the room, while the Exit IR Sensor detects people leaving.

Whenever the Entry IR Sensor is triggered, the Arduino Uno increments the occupancy count. Similarly, when the Exit IR Sensor is triggered, the occupancy count is decremented. The updated occupancy count is displayed on the Arduino Serial Monitor.

The Arduino controls the 8-channel relay module based on the current occupancy count. As the occupancy count increases, additional relay channels are activated, and the corresponding onboard LEDs illuminate. When the occupancy count decreases, the relays are deactivated accordingly.

Note: In this prototype, the occupancy count is limited to a maximum of **eight** to match the available 8-channel relay module used for demonstration. This limit is introduced solely for visualization purposes and does not represent a limitation of the occupancy detection system. The occupancy counting algorithm can be extended to support a larger number of occupants by modifying the software and using suitable output hardware.

9. Circuit Diagram



Note: Two 4-channel relay modules are shown in the circuit diagram to represent the connections of the 8-channel relay module used in the hardware prototype, as an equivalent 8-channel component was not available in the design software.

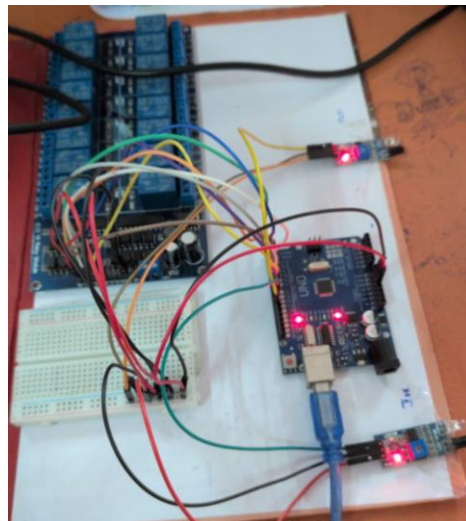
The circuit diagram illustrates the electrical connections between the Arduino Uno, the two IR sensors, the 8-channel relay module, and the power distribution through the breadboard.

The Arduino supplies power to the sensors and relay module through the breadboard. The output pins of the two IR sensors are connected to the Arduino input pins for entry

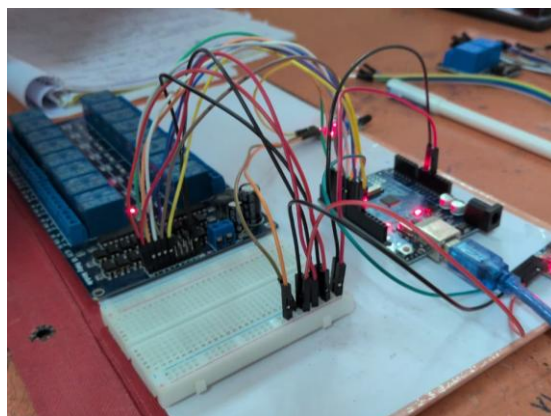
and exit detection, while the relay module input pins are connected to the Arduino digital output pins.

10. Hardware Prototype

The hardware prototype was assembled using an Arduino Uno, two IR sensors, an 8-channel relay module, a breadboard, and jumper wires. The Arduino receives signals from the IR sensors and controls the relay module based on the occupancy count. During testing, the onboard LEDs of the relay module were used to indicate the active relay channels corresponding to the occupancy count.



Front View



Angled View

11. Source Code

```
// ----- Pin Definitions -----  
  
const int relayPins[8] = {2, 3, 4, 5, 6, 7, 8, 9};  
  
const int INpin = 10;  
  
const int OUTpin = 11;  
  
// ----- Variables -----  
  
int COUNT = 0;  
  
int currentIN = HIGH;  
  
int previousIN = HIGH;  
  
int currentOUT = HIGH;  
  
int previousOUT = HIGH;  
  
// ----- Setup -----  
  
void setup()  
{  
    Serial.begin(9600);  
  
    pinMode(INpin, INPUT);  
  
    pinMode(OUTpin, INPUT);  
  
    for (int i = 0; i < 8; i++)  
    {  
        pinMode(relayPins[i], OUTPUT);  
  
        digitalWrite(relayPins[i], HIGH); // Relay OFF (Active LOW)  
    }  
  
    Serial.println("Occupancy Monitoring System Started");  
}
```



```

// ----- Loop -----
void loop()
{
    currentIN = digitalRead(INpin);
    currentOUT = digitalRead(OUTpin);

    // ----- Person Entered -----
    if (previousIN == HIGH && currentIN == LOW)
    {
        // Limit set to 8 to match the 8-channel relay module used in this prototype.
        if (COUNT < 8)
            COUNT++;

        Serial.println("-----");
        Serial.println("Person Entered");
        Serial.print("Occupancy = ");
        Serial.println(COUNT);
        updateRelays();
        delay(300);    // Debounce
    }

    // ----- Person Exited -----
    if (previousOUT == HIGH && currentOUT == LOW)
    {
        if (COUNT > 0)
            COUNT--;

        Serial.println("-----");
    }
}

```

```

    Serial.println("Person Exited");

    Serial.print("Occupancy = ");

    Serial.println(COUNT);

    updateRelays();

    delay(300);    // Debounce
}

previousIN = currentIN;

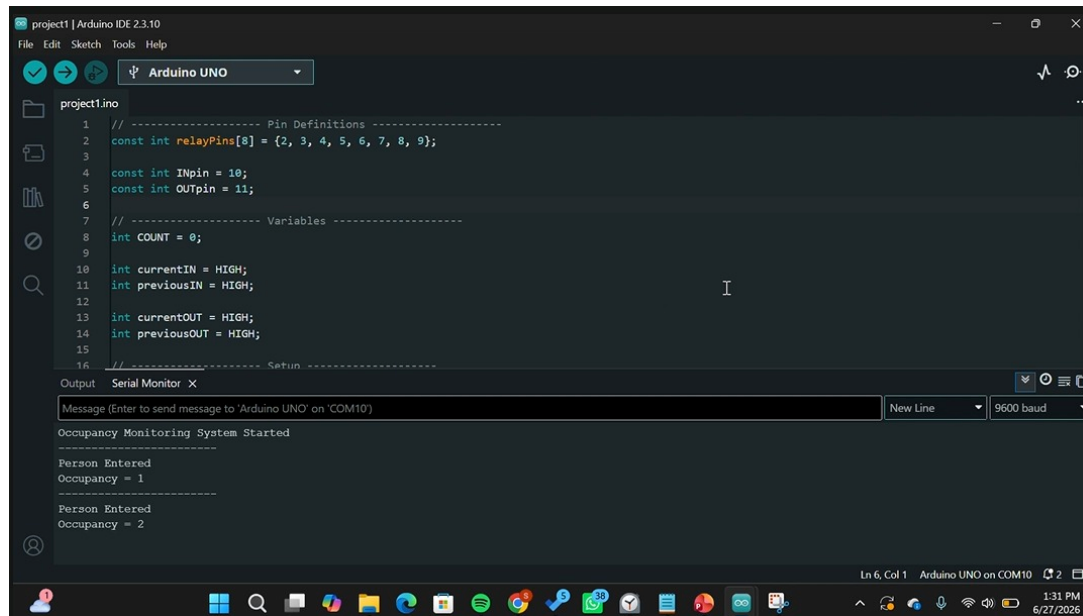
previousOUT = currentOUT;
}

// ----- Relay Control -----

void updateRelays()
{
    for (int i = 0; i < 8; i++)
    {
        if (i < COUNT)
            digitalWrite(relayPins[i], LOW);    // Relay ON
        else
            digitalWrite(relayPins[i], HIGH);    // Relay OFF
    }
}

```

12. Working of the Prototype

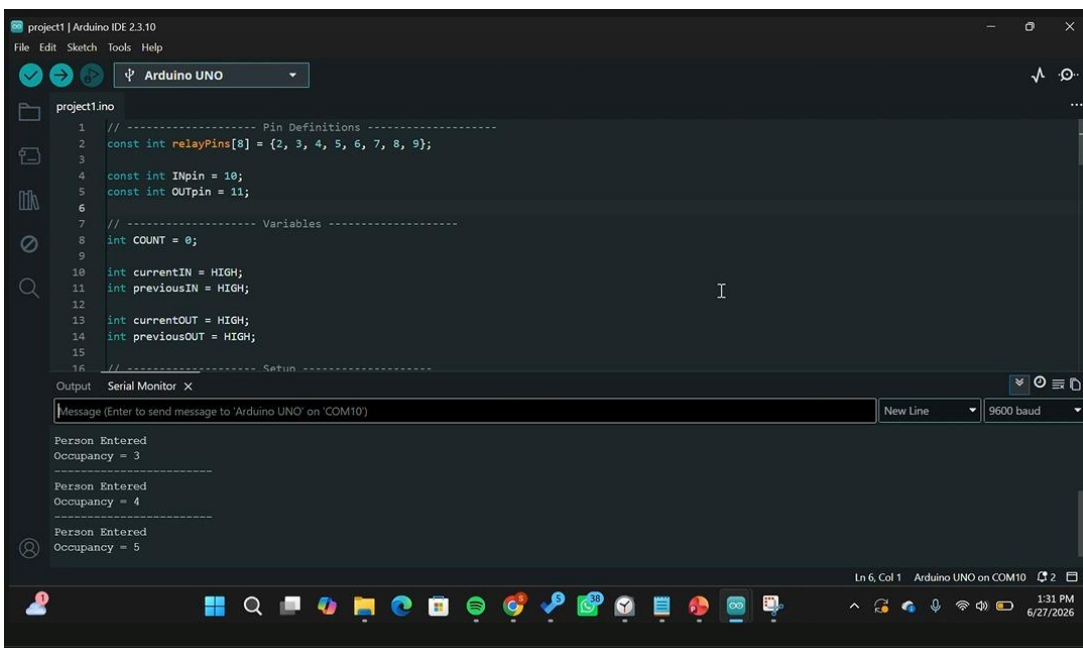


The screenshot shows the Arduino IDE interface with the 'project1.ino' file open. The code in the editor includes pin definitions, variable declarations, and initialization. The Serial Monitor at the bottom displays the output of the program, showing the initial state of the occupancy monitoring system.

```
1 // ----- Pin Definitions -----
2 const int relayPins[8] = {2, 3, 4, 5, 6, 7, 8, 9};
3
4 const int INpin = 10;
5 const int OUTpin = 11;
6
7 // ----- Variables -----
8 int COUNT = 0;
9
10 int currentIN = HIGH;
11 int previousIN = HIGH;
12
13 int currentOUT = HIGH;
14 int previousOUT = HIGH;
15
16 // ----- Setup -----
```

Serial Monitor Output:

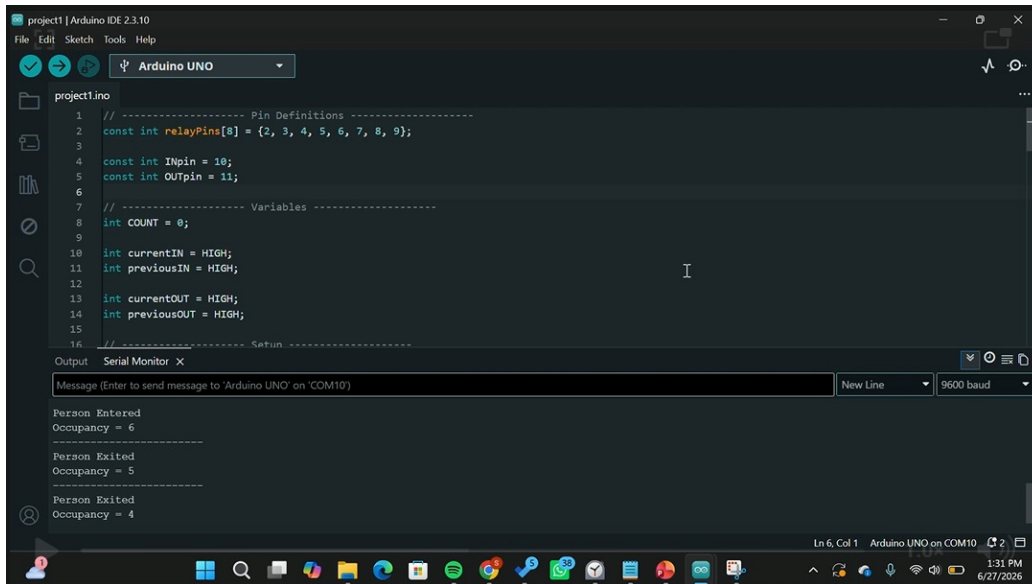
```
Occupancy Monitoring System Started
-----
Person Entered
Occupancy = 1
-----
Person Entered
Occupancy = 2
```



This screenshot shows the same Arduino IDE interface as the previous one, but the Serial Monitor now displays three more lines of output, indicating that three more people have entered the system.

```
Person Entered
Occupancy = 3
-----
Person Entered
Occupancy = 4
-----
Person Entered
Occupancy = 5
```

PERSON ENTERING



PERSON EXITING

13. Results and Discussion

The proposed Real-Time Occupancy Detection System was successfully implemented and tested using an Arduino Uno, two IR sensors, and an 8-channel relay module. During testing, the Entry IR Sensor correctly increased the occupancy count, while the Exit IR Sensor decreased the occupancy count. The updated occupancy count was displayed on the Arduino Serial Monitor in real time.

The Arduino controlled the relay module according to the occupancy count. As the occupancy increased, additional relay channels were activated and the corresponding onboard LEDs illuminated. When the occupancy decreased, the relay channels were deactivated accordingly.

The prototype successfully demonstrated occupancy counting from **0 to 8**. The maximum count of eight was intentionally selected to match the available 8-channel relay module and provide a clear visual representation of occupancy. This implementation choice does not limit the occupancy detection algorithm, which can be extended to support a larger number of occupants using suitable hardware modifications.

14. Applications

The proposed system can be used in various real-world applications, including:

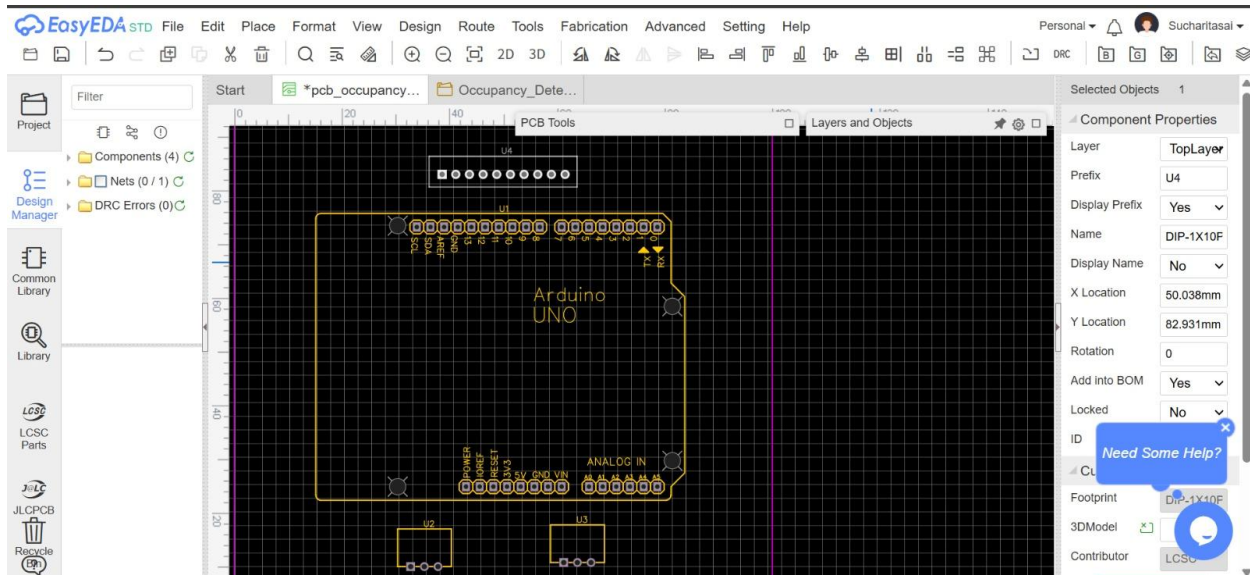
- Smart homes
- Offices and workplaces
- Classrooms and laboratories
- Conference rooms
- Libraries
- Commercial buildings
- Energy management systems
- Building automation
- Public occupancy monitoring

15. Future Scope

The proposed system can be further enhanced by incorporating additional features to improve functionality and scalability.

- Integrate IoT platforms for remote occupancy monitoring.
- Store occupancy data on cloud servers for analysis.
- Develop a mobile application for monitoring occupancy.
- Connect the relay module to actual electrical appliances such as lights and fans.
- Extend the occupancy counting capability beyond eight users by modifying the software and using suitable output hardware.
- Integrate the system with security and access control systems.

16. PCB Layout and Gerber Files



17. Demo Video of the Prototype

https://drive.google.com/file/d/1zHfr7VFqM12LqBdDJ3e7b-XL28vh8_v8/view?usp=sharing

18. Conclusion

The Real-Time Occupancy Detection System was successfully designed and implemented using an Arduino Uno, two IR sensors, and an 8-channel relay module. The system accurately maintained the occupancy count based on entry and exit events while providing real-time monitoring through the Serial Monitor.

The relay module demonstrated occupancy-based automation by activating relay channels according to the occupancy count. The prototype proved to be simple, reliable, and cost-effective, making it suitable for small-scale occupancy monitoring applications. The proposed system also provides a foundation for future IoT-enabled smart building and energy management solutions.

19. References

- Arduino Uno Rev3 Technical Specifications, Arduino Documentation.
- Arduino IDE Documentation and User Guide, Arduino Official Documentation.
- ATmega328P Microcontroller Datasheet, Microchip Technology Inc.
- Infrared (IR) Obstacle Avoidance Sensor Module Datasheet, Electronics Components Manufacturer Documentation.
- Simon Monk, *Programming Arduino: Getting Started with Sketches*, McGraw-Hill Education.